Kai Ling

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PROFILE

Computer Science PhD candidate at Purdue University developing Transformer and diffusion models, training-data pipelines, and reproducible evaluation systems. Experienced in multi-node GPU training and memory optimization, with full-stack experience deploying a public research application.

EDUCATION

Purdue University — **West Lafayette, USA**

PhD in Computer Science, expected May 2027 — Aug. 2019 – Present

Master of Science in Computer Science — May 2025

Huazhong University of Science and Technology (HUST) — **Wuhan, China**

Bachelor of Science in Computer Science and Technology — Aug. 2015 – July 2019

GPA: 3.8/4.0

SELECTED WORK & RESEARCH EXPERIENCES

Research Assistant | Kihara Lab | Purdue — **West Lafayette, USA**

Advisor: Daisuke Kihara, Professor, Purdue University

Lead Developer | NuFold Multimer: Generative Modeling for RNA Complexes — Jan. 2026 – Present

- Extended a Transformer-based RNA structure model from single-chain to multi-chain prediction, integrating a diffusion module to generate multiple candidate 3D structures.
- Built a data-curation and teacher-inference pipeline, running inference on approximately 20,000 RNA complexes; integrated multiple data sources, corrected inconsistent labels and identifiers, and applied confidence-based filtering for training.
- Ran matched training ablations using 8 NVIDIA A100 80GB GPUs across 4 nodes to evaluate diffusion-decoder stability.
- Enabled training crops of approximately 1,000 tokens by integrating NVIDIA cuEquivariance and a memory-efficient diffusion implementation.
- Benchmarked AlphaFold3 and Boltz-2 on RNA complexes to compare prediction accuracy and characterize model limitations.

Evaluation & Benchmarking | CryoASTRA: Cryo-EM Atomic Structure Modeling — July 2026 –Sept. 2026

- Built a Python/SLURM benchmarking pipeline for CryoASTRA and seven baselines across 103 test cases, with resumable execution, validated result caching, and input/code provenance tracking.
- Standardized prediction-to-reference matching and coverage-aware metrics for incomplete outputs; implemented matched-target comparisons and bootstrap confidence intervals to quantify performance differences.
- Evaluated five generated samples per input; benchmark results showed a mean structural similarity score of 0.972 (TM-score) and 92.5% atom-placement accuracy within 3 Å across 101 protein-containing targets, with a statistically significant TM-score gain of 0.117 over CryoAtom2.

Lead Developer | Distance-AF Multimer: Constraint-Guided Protein Complex Modeling — Feb. 2024 – Sept. 2025

- Implemented per-target test-time optimization to adapt pretrained model predictions to user-supplied geometric constraints without retraining the underlying model.
- Designed a coarse-to-fine optimization strategy to improve convergence from inaccurate initial predictions and reduce structural artifacts.
- Built a PyTorch optimization workflow supporting structures with 50,000+ atoms, parallelizing independent target optimizations across GPUs.

- Evaluated on 27 challenging multimer targets, 8 peptide complexes, and 6 assemblies with 5+ chains; achieved lower coordinate error (RMSD) than Chai-1, Protenix, and AlphaLink2 on 22/26, 19/27, and 26/27 benchmark targets, respectively, with up to 57.6 Å lower RMSD than AlphaFold-Multimer.
- Conducted ablations across 11 complexes with 0–30 distance constraints and uncertainty of ±2.5 Å and ±5 Å; retained acceptable docking accuracy on 8/11 targets under both uncertainty settings.

First-author preprint on bioRxiv (2026).

Designer & Sole Developer | DAQ-Refine: Cryo-EM Protein Model Refinement Service Aug. 2023 – Jan. 2024

- Independently built and deployed a public React/Flask application with interactive Mol* visualization; integrated into EMSuite and used for approximately 10–20 refinement jobs per month.
- Migrated a Colab workflow to lab infrastructure and automated SLURM job orchestration to run three refinement strategies per protein chain, select the best results, and assemble the final structure.
- Developed reusable input-validation, early error-reporting, and progress-tracking components for integration with other EM-server algorithms.
- Diagnosed recurring job failures reported by lab researchers and external users; added optional sequence alignment, specific input-error messages, and input-preparation documentation to help users resolve incompatible inputs.

Research Assistant | MSSN Lab | Purdue **West Lafayette, USA**

Advisor: Chunyi Peng, Professor, Purdue University

- Independently developed the initial mperf platform using Django and Android for remote experiment control and automated cellular-network measurements; collected 705 hours of data comprising 4.6 million throughput samples across three U.S. carriers, supporting two published studies.
- Redesigned MMMap's database and visualization workflows for 32,000+ network cells and 18,700+ handoff events, achieving a 10× speedup in data analysis and display.
- Collected and analyzed 45 hours of UAV network measurements over 1,010 km, using radio-access-network traces to compare airborne and ground-level throughput, latency, and mobility failures.
- Applied lightweight formal analysis to 3GPP Release 17 IoT protocols, identifying four vulnerabilities and developing 12 proof-of-concept demonstrations.

PUBLICATIONS(SELECTED)

- **Ling, K.** et al. Distance-AF Multimer: A Multi-Step Approach for Protein Complex Structure Modeling with User-Defined Distance Constraints. *bioRxiv*, 2026.
- **Ling, K.** et al. Peptide–protein docking: from physics-based models to generative intelligence. *Chemical Communications*, 2026.
- Li, Y. et al. (incl. **Ling, K.**). Experience: A Five-Year Retrospective of MobileInsight. *ACM MobiCom 2021*.
- Deng, H., **Ling, K.** et al. Unveiling the Missed 4.5G Performance in the Wild. *ACM HotMobile 2020*.
- **Ling, K.*** et al. MMMap: Mobility Management Map of Global Carriers Networks At Your Hands. *ACM IMC 2018* (Poster, *co-first).

TEACHING EXPERIENCE

Teaching Assistant | CS536: Computer Networking | Purdue 2020 – 2025(selected semesters)

- Led office hours and debugging sessions, authored assignments and exam questions, and graded across multiple cohorts — explaining complex networking systems and protocols to students.

SKILLS

- **Programming Languages**: Python, C++, Java, SQL, Bash.
- **ML / DL:** PyTorch, Transformers, diffusion models, distributed multi-GPU training, GPU inference, training with teacher-generated data, model evaluation, NVIDIA cuEquivariance.
- **Infrastructure:** Linux, HPC / SLURM, Docker, Git, CI/CD.
- **LLM / GenAI:** OpenAI and Anthropic APIs, RAG, FAISS, pgvector, function calling, agents.
- **Application Development:** React, Flask, Django, REST APIs, Mol*.